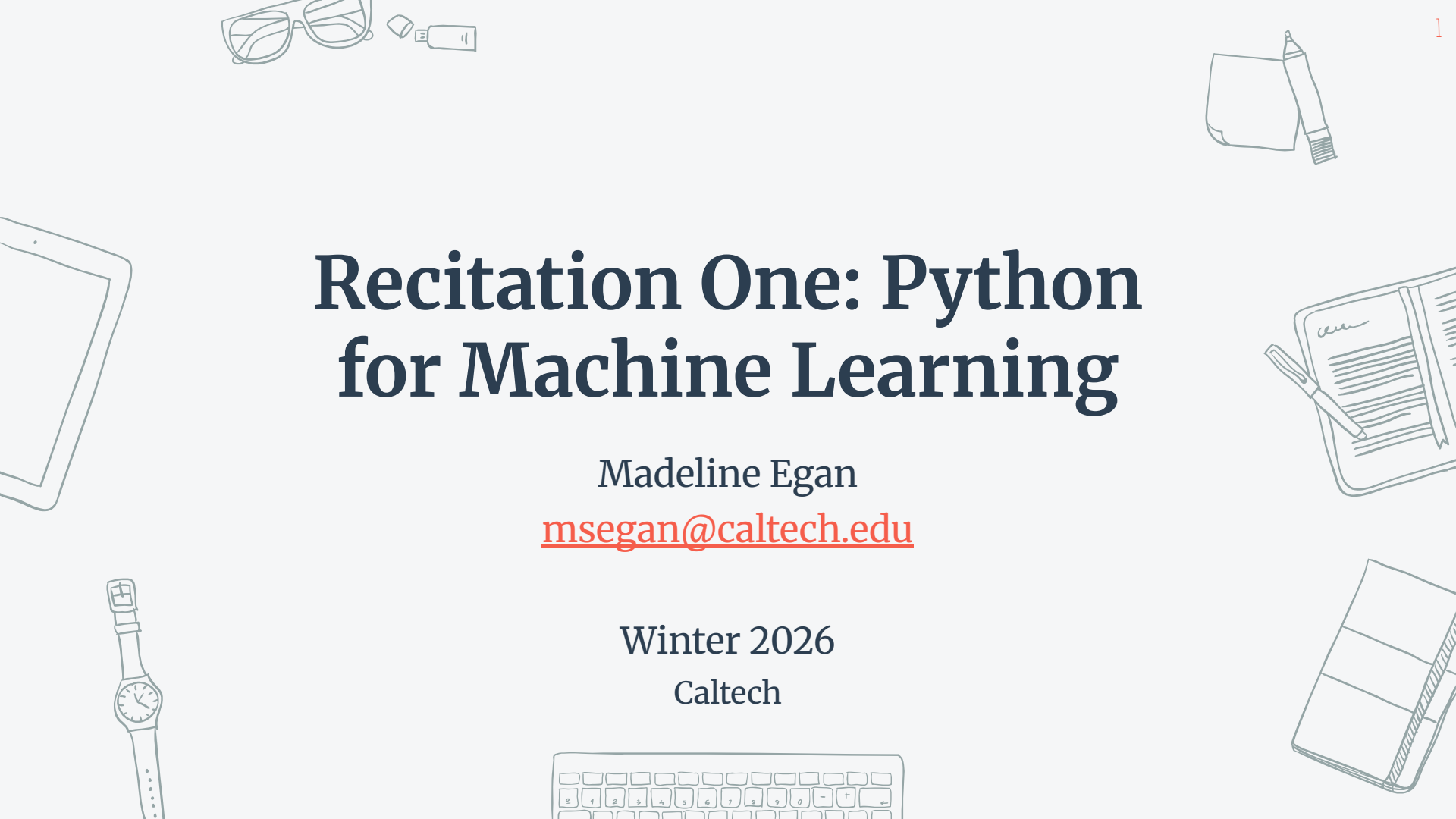




Recitation One: Python for Machine Learning

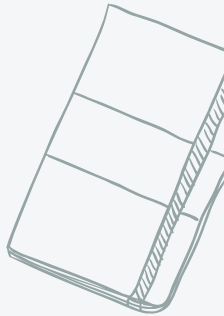
Madeline Egan

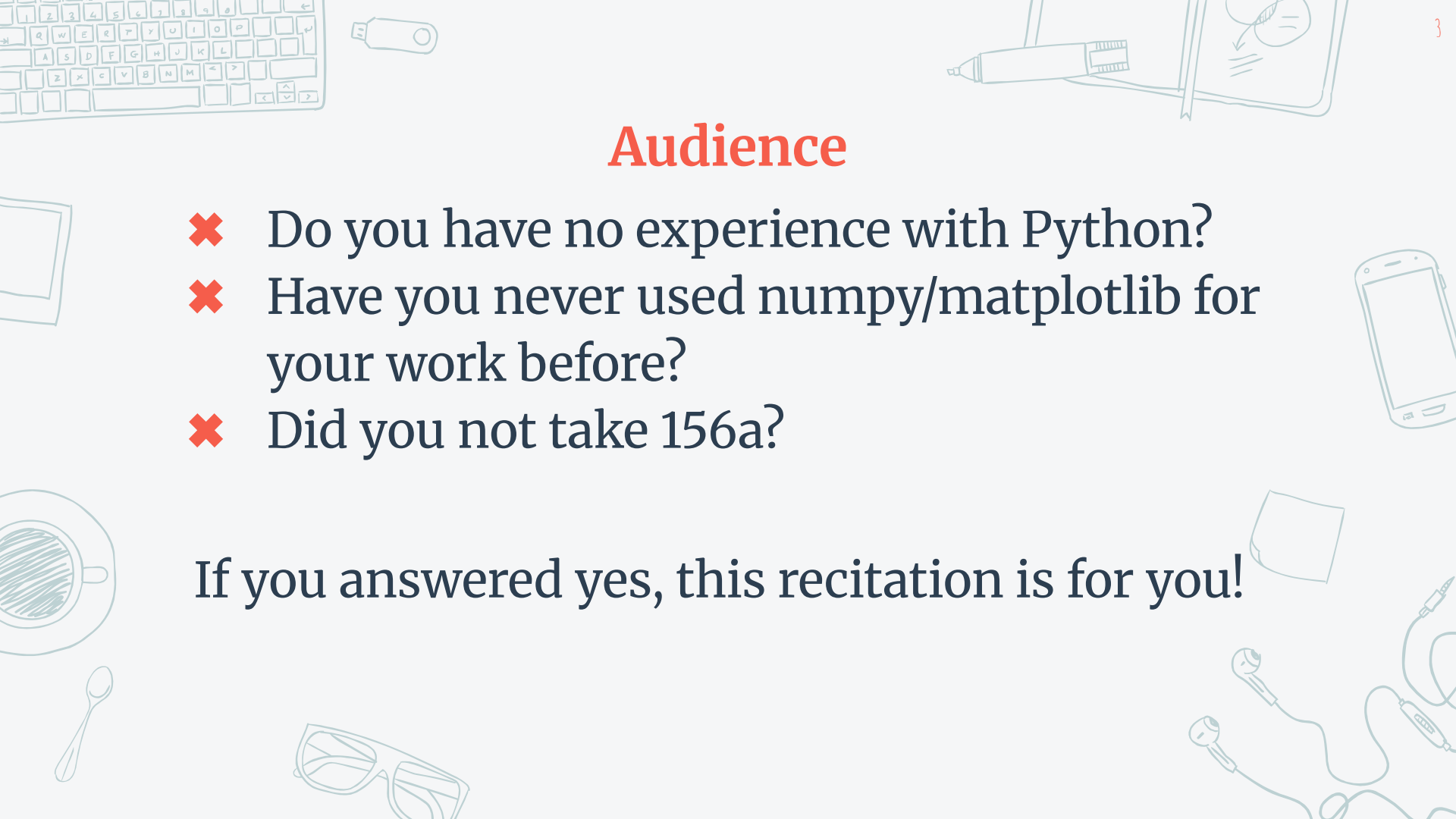
mseган@caltech.edu

Winter 2026

Caltech

A quick check

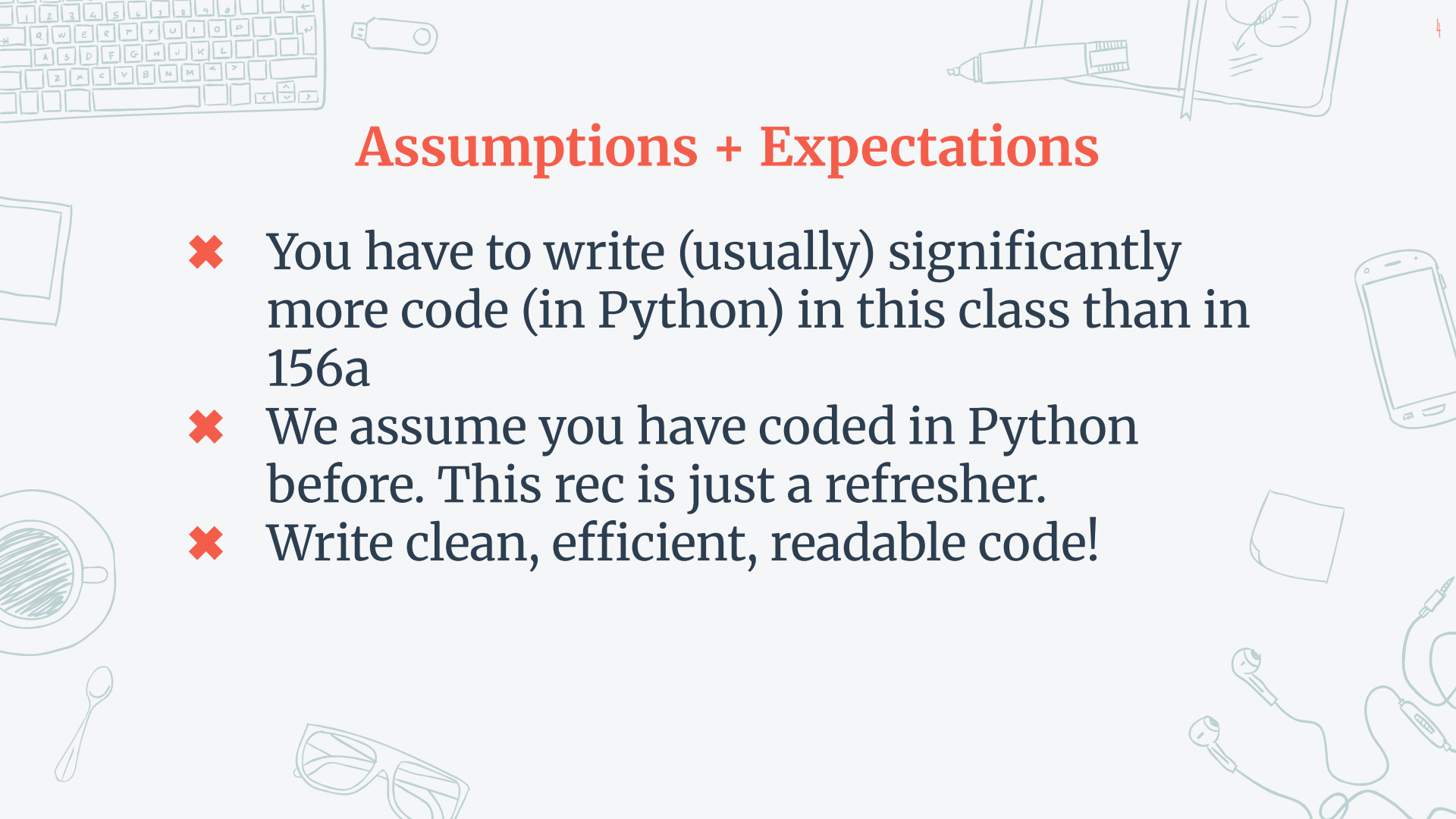




Audience

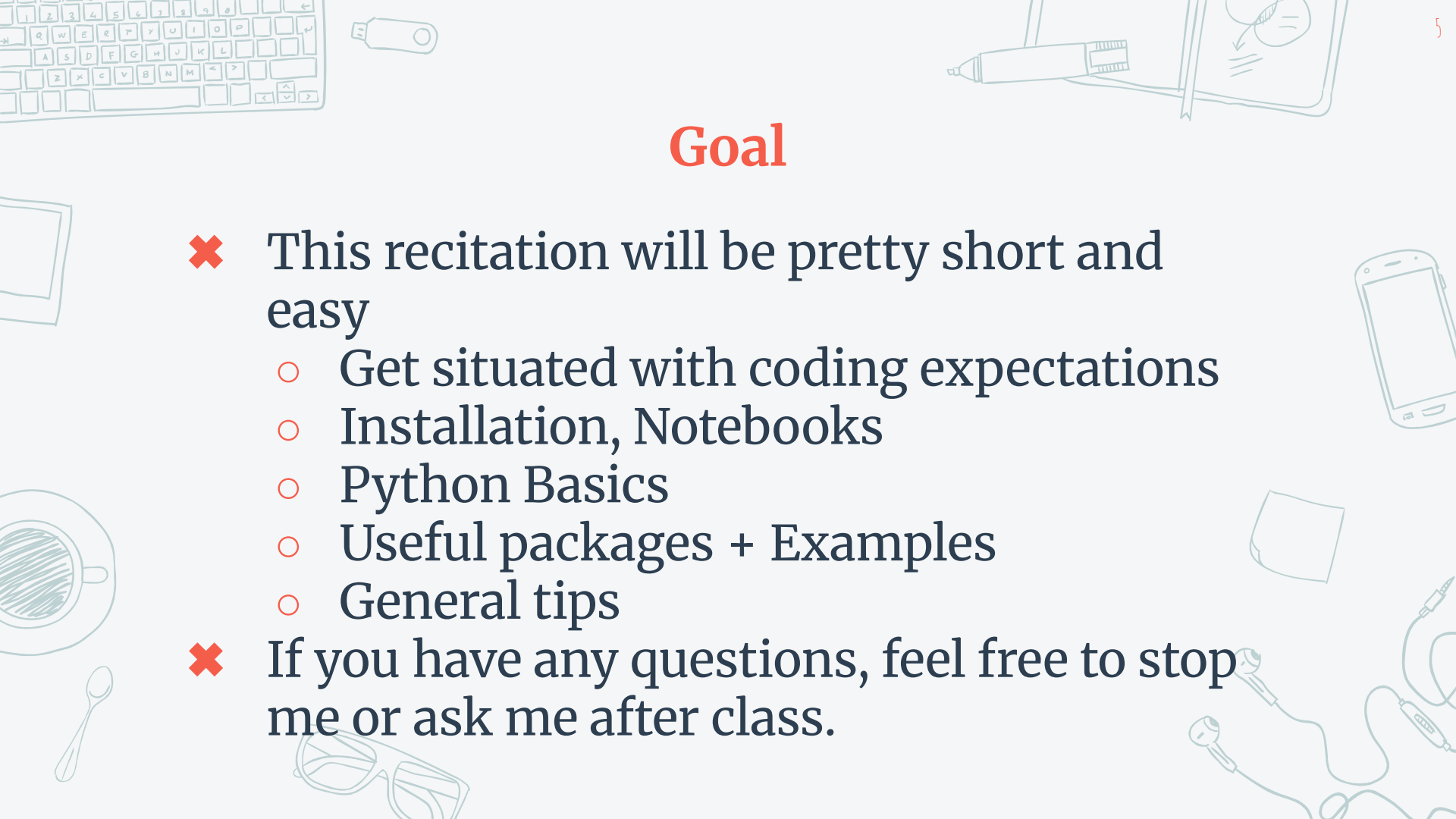
- ❌ Do you have no experience with Python?
- ❌ Have you never used numpy/matplotlib for your work before?
- ❌ Did you not take 156a?

If you answered yes, this recitation is for you!



Assumptions + Expectations

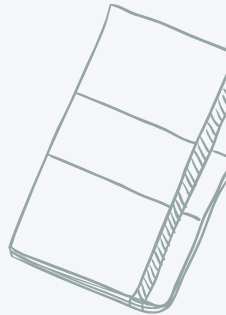
- ✖ You have to write (usually) significantly more code (in Python) in this class than in 156a
- ✖ We assume you have coded in Python before. This rec is just a refresher.
- ✖ Write clean, efficient, readable code!



Goal

- ✖ This recitation will be pretty short and easy
 - Get situated with coding expectations
 - Installation, Notebooks
 - Python Basics
 - Useful packages + Examples
 - General tips
- ✖ If you have any questions, feel free to stop me or ask me after class.

Setup



Installing Python

✖ Python 3

- <https://www.python.org/downloads/>

✖ Good idea to have Python locally

Jupyter Notebooks

- ✖ You can install with pip or use anaconda:
 - `pip install notebook`
 - <https://jupyter.readthedocs.io/en/latest/install.html>
- ✖ It comes with anaconda
 - <https://www.anaconda.com/>
- ✖ Lets you work with notebooks locally.



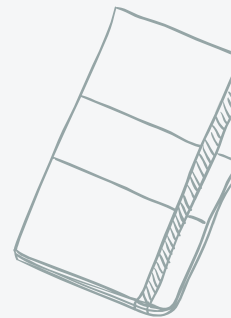
Visual Studio Code

- ❌ VS code allows you to live share
- ❌ You can still use Jupyter notebook (.ipynb)
- ❌ Less glitchy than Google colab when collaborating on mini projects

Google Colab Notebooks

- ✖ Modify your notebook online
- ✖ Download in .ipynb format
- ✖ Excellent for writing code incrementally and testing as you go
- ✖ <https://colab.research.google.com/>

Python



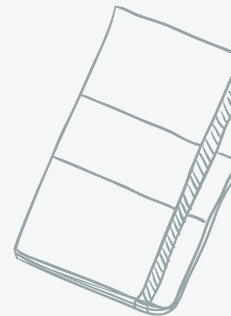
Python Basics

- ✖ Syntax
- ✖ Printing
- ✖ Loops
- ✖ Functions
- ✖ Clean Code

Python Resources

- ✖ Python cheat sheet:
www.pythoncheatsheet.org
- ✖ Python for Data Science - A Cheat Sheet for Beginners:
<https://www.datacamp.com/community/tutorials/python-data-science-cheat-sheet-basics>

Packages



Packages

- ✖ You can use a package manager:
 - <https://www.anaconda.com/download>
- ✖ Or you can use pip: `pip install numpy`
- ✖ You need
 - `numpy`
 - `scikit-learn`
 - `matplotlib`
- ✖ Using git and LaTeX is recommended :)

numpy

- ✗ Used for numerical computing/matrix operations
- ✗ Your data is going to be in a matrix, so manipulate it with numpy

numpy Resources

- ✖ **Python numpy Tutorial:**
<https://cs231n.github.io/python-numpy-tutorial/>
- ✖ **numpy user guide:**
<https://numpy.org/doc/stable/user/index.html>

scikit-learn

- ✖ Used for basic ML algorithms, tools and techniques
- ✖ User guide:
https://scikit-learn.org/stable/user_guide.html

K-fold cross validation

- ❌ How to pick a model set?
- ❌ Approximate a generalization error
- ❌ Idea: validation sets

K-fold cross validation

- ✗ Algorithm: k fold cross validation
- ✗ For 1, 2, ... k
 - Use the first k th of data as validation, and train on the remaining data points
 - Evaluate error on the validation set, and store it
- ✗ Average the validation errors and return this as the k -fold cross validation error
- ✗ How to implement this? Not hard but it requires work!

sklearn.model_selection.KFold

```
class sklearn.model_selection.KFold(n_splits=5, *, shuffle=False, random_state=None)
```

[\[source\]](#)

K-Folds cross-validator

Provides train/test indices to split data in train/test sets. Split dataset into k consecutive folds (without shuffling by default).

Each fold is then used once as a validation while the k - 1 remaining folds form the training set.

Read more in the [User Guide](#).

Parameters:

n_splits : int, default=5

Number of folds. Must be at least 2.

Changed in version 0.22: `n_splits` default value changed from 3 to 5.

shuffle : bool, default=False

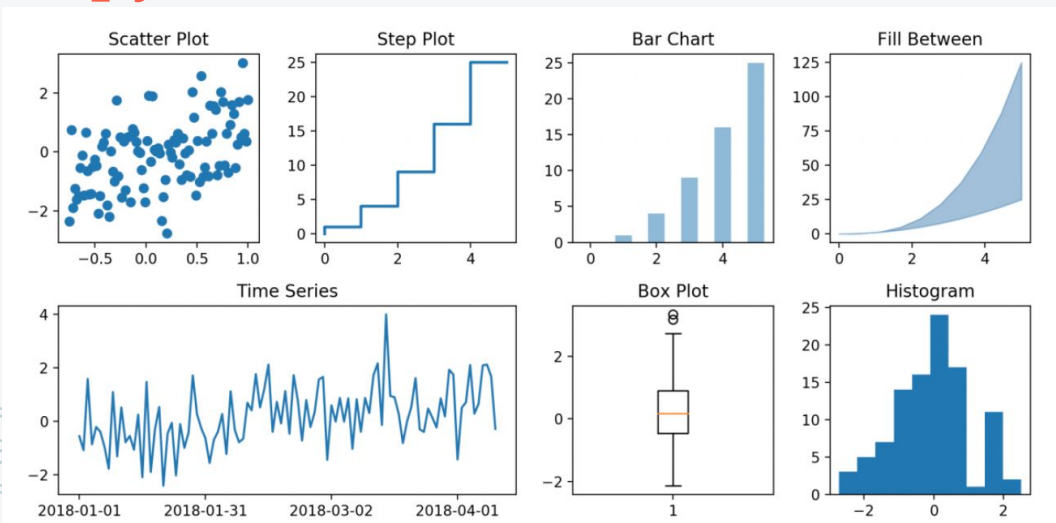
Whether to shuffle the data before splitting into batches. Note that the samples within each split will not be shuffled.

random_state : int, RandomState instance or None, default=None

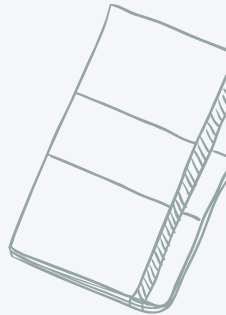
When `shuffle` is True, `random_state` affects the ordering of the indices, which controls the randomness of each fold. Otherwise, this parameter has no effect. Pass an int for reproducible output across multiple function calls. See [Glossary](#).

matplotlib

- ❌ Use it to plot things
- ❌ You'll probably need to make a plot on every set!
- ❌ Guide: <https://pythonguides.com/matplotlib-in-python/>



Tips for Success



The Supervised Learning Recipe

- ✖ Get training data
- ✖ Pick a model class
- ✖ Pick a loss function
- ✖ Pick a learning objective to optimize

Debugging Tips

- ✖ Print it
- ✖ Google it
- ✖ Try using dummy data
- ✖ Ask for help!
- ✖ Take a walk
- ✖ Take a nap

Coding Resources

- ❌ Remember: Stack Overflow is your best friend!
- ❌ Python cheat sheet: www.pythoncheatsheet.org
- ❌ Python for Data Science - A Cheat Sheet for Beginners:
<https://www.datacamp.com/community/tutorials/python-data-science-cheat-sheet-basics>
- ❌ Scikit User guide:
https://scikit-learn.org/stable/user_guide.html
- ❌ Python numpy Tutorial:
<https://cs231n.github.io/python-numpy-tutorial/>
- ❌ numpy user guide:
<https://numpy.org/doc/stable/user/index.html>